

LIVING IN THE
ANTHROPOCENE

THE AGE OF HUMANS

The Moral Dilemma We Face in the Age of Humans

Humans are proficient problem solvers—but so far that trait has come at a cost. Can our species remain resilient without destroying the world?



Our fundamentally human social, ecological, and behavioral adaptations have, over time, ratcheted up our adaptability. (Karen Carr Studios/Human Origins Program/SI)

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Six million years ago, an ancestor, standing upright, began the human venture. This upright predecessor evolved near the Equator, in Africa, followed by six million years of experiments in being human, as new species originated, diversified and became extinct.

From the outset, this venture took place in an era of instability and uncertainty. Although we used to think of East Africa as the nurturing “cradle of mankind,” it’s now understood to have created precarious tests of survival and adaptability. “The cauldron of human evolution” is a phrase I prefer, reflecting the roiling events and churning process that defined the thin line between thriving and decline, between survival and extinction in the era of human origins.

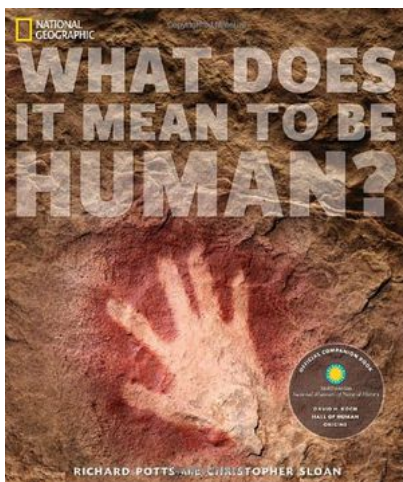
After several million years, the sole survivor of this radiation of bipedal species is *Homo sapiens*, worldwide in its extent: a turning point in the history of life due to our capacity to alter the world.

We see this among the oldest [artifacts](#) in the Smithsonian collections: a chipped stone, some sharp stone flakes and a hammerstone, comprising the oldest toolkit, made by human ancestors two million years ago. They are simple things, yet they mark the first stages of our species’ ability to alter and rearrange the surroundings.

The chipped stone shown below was carried eight miles from its source to a place where antelopes and young zebra were butchered, and their bones broken to extract nutritious marrow, and tubers were dug out of the ground. The hammerstone could crush more effectively than an elephant’s molar. A sharp edge could cut more efficiently than a lion’s canine, or sharpen a stick that could dig with greater yield than a warthog’s tusk to get tubers and roots under the ground. In fact, any food that could be eaten by a large omnivorous mammal could be obtained with these simple modifications of the environment.



Among the oldest artifacts in the Smithsonian collections are a chipped stone, some sharp stone flakes and a hammerstone, comprising the oldest toolkit, made by human ancestors two million years ago. (Smithsonian Institution)



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This earliest ability to alter things took place during a time of strong fluctuation in East African climate and landscapes. The adaptability afforded by these strange behaviors—chipping stones and moving rocks from one place on the landscape to another—proved to be of such survival value in this ever-shifting era that technology stayed around and developed, and it became part of the foundation of human life.

The first two-thirds of our evolutionary history were exclusively in Africa. Shortly after 2 million years ago, our genus, the genus *Homo* (which is African in origin), began to spread to new places, taking with it its ability to modify things. With the species *Homo erectus*, the ability to explore and disperse to new places enabled it to endure. In fact, *Homo erectus* survived nine times longer than our own species has been around so far, and from *erectus* we inherited a propensity to explore and colonize.

Yet much of what is distinctive of our species evolved later, over the past million years: attaining a particularly large brain relative to body size; controlling fire and making shelters indicative of a central place on the landscape where the social group returned "home" in a way familiar to humans today; and prolonging the pace at which we grow up, with enormous implications for the time, care and energy we put into raising children—as well as for learning and the capacity for culture.

By 300,000 years ago, as documented by our recent excavations in the [Kenya Rift Valley](#), we see the first obvious clues that a transition had occurred toward innovation: tool kits that were new, including sharpened projectiles; pigments that could be used for coloring, emblematic of an increasingly complex ability to use symbols and language; the development of social networks and the exchange of resources among groups living far apart; and, eventually, the diversifying of cultures, which multiplied the options of our species, diverse expressions of what it means to be human.

These aspects of our heritage arose in a dynamic, unpredictable world. On a global scale, the past 6 million years have comprised one of the most dramatic periods of climate oscillation and environmental instability of the Cenozoic Era (covering the past 65 million years). Every paleoclimate and paleoenvironmental record studied over the past 40 years has two signals—the overall trend and the amplitude of variability. Up until about 20 years ago, every student of human origins considered the variability as noise in the all-important trend toward a cooler, drier Earth: for example, the development of savanna grasslands in Africa, and Ice Age

conditions in northern latitudes. The direction of climate change—and the onset of a particular ancestral habitat—was thought to be the key signal that elicited the development of uniquely human adaptations.



A sharp edge could cut more efficiently than a lion's canine, or sharpen a stick that could dig with greater yield than a warthog's tusk to get tubers and roots under the ground. (Smithsonian Institution)

Yet all of those many dozens of environmental records show evidence of dramatic instability between wet and arid, between cool and warm. As a result, variability and uncertainty have become the new theme in the environmental story of human origins.

The overarching narrative of human evolution has thus significantly changed. It has changed from a story of how the human lineage came to have dominion over its ancestral environment, to a story of evolving adaptability and persistent change in the challenges to survival.

There are many conclusions that can be drawn from these new perspectives on human origins. During the era of human evolution, the natural world has had no enduring, stable baseline. Over the past several million years, high rates of extinction have occurred in most groups of vertebrates. This is true even in our own evolutionary group. Out of a minimum of 18 different species of evolutionary ancestors and cousins, only one lineage—our species—has survived. All the other ways of life of earlier hominins have gone extinct, even though each species possessed at least some of the unique distinguishing characteristics of human life.

The difference between humans today and our extinct immediate relatives (in the evolutionary sense) is that our basic adaptations rely heavily upon an ability to alter the surroundings. This is our mode of survival.

Our species *Homo sapiens* possesses, through natural evolutionary heritage, an extraordinary capacity to modify landscapes; the distribution of food, water and other resources; and, most interesting, ourselves. We have an unprecedented proclivity to alter our ways of life, our systems of belief and our transactions with one another and the world around us. This is responsible for the vast diversity of human behavior and our species' cultural diversity.



In East Africa, a male *Paranthropus boisei* tugs at root. (Bronze artist: John Gurche)

From the standpoint of human origins, the starting point in thinking about the Age of Humans or the Anthropocene is that we live in the world by altering it.

Our fundamentally human social, ecological and behavioral adaptations have, over time, ratcheted up our adaptability—the ability to manage the immediate world, to cushion the unpredictable, to survive novelty—all through an extraordinary ability to alter the surroundings. We buffer uncertainty and instability by changing how the world is.

Now we find ourselves where the planetary scale of human impact is unquestionable.

Decreased biodiversity, revised biogeochemical cycles, and novel combinations of climatic and ecological conditions arise from the existence of people everywhere. Covering more than 50 percent of today's land surface are human-dominated ecosystems, where energy flow is channeled largely toward human needs. When you add up the areas humans occupy, use, or destroy, the total comes to some 83 percent of the Earth's viable land surface. About six times more water is controlled by dams or in manmade lakes than occurs as free-flowing water on the continents. Regarding the atmosphere, despite differences in opinion over the exact future of rising CO₂, what seems to get lost in the noise of manufactured debate is that even the most conservative estimates of sea-level rise would eventually inundate areas occupied by ten percent of the human population. Given the sharp rise in population during our lifetimes, the implications of 700 to 900 million people displaced, or requiring new livelihoods, have hardly been imagined.

Critical to imagining life in the Anthropocene is the importance of narrative in continually reshaping ourselves. We—and our altering tendencies—are embedded in a very dynamic natural world and fully interconnected with it. Revising the entwined human-and-nature narrative to reflect this point is, I believe, essential in how we will shape the future.



Within the past 100,000 years, *Homo floresiensis* struggled for survival on a small Indonesian island. (Bronze artist: John Gurche)

The Age of Humans, or the concept of the Anthropocene, has largely gained traction by drawing attention to the harm we cause. There is justifiable hand-wringing over the unintended consequences of human activity—the “downstream effects” of human decisions, waste and emissions, the self-interest in the use of land and resources that underpin livelihoods and personal security and comforts.

I have begun to imagine what it would be like to have a different starting point in discussing this Age of Humans—one where we envision intended and purposeful consequences. What will it take to shape a world that is positive, meaningful, beneficial to life, in general, and to human welfare?

There is a young man who lives where I work in the Rift Valley of East Africa. He is the son of a very old man who owns land bordering a river dry most of the year, except during the rains. Over the last several years, this young man has decided to cut the trees in the places where his father can no longer visit, including all the trees along the river, the trees that hold up the river bank.

The people of the community have a great depth of knowledge about landscapes, the care of their livestock, and the responsible nurturing of environment and wildlife. Yet the decision of this young man to burn trees, make charcoal and make money by selling the charcoal for personal gain, are, during the rains, causing the loose silt from the river bank to erode and wash downstream. The sediment is filling the livestock watering holes used by the entire community. Soon, the managed pools will be filled, they will be gone, and so will the water for the cattle and the wildlife. Literally, this is a downstream effect.

Everyone in the community knows this man and the impact of his supposedly hidden enterprise. But they have no idea what to do. They ask: Shouldn't a person have rights to what is his—on his land, with his trees? This is a problem of the Anthropocene.



This female *Homo erectus* butchered animals using simple stone tools. Bronze artist: John Gurche (Bronze artist: John Gurche)

This story underlines the fact that the young man's decisions are part of the community; in a way, they belong to the community. His actions have connected him with all his neighbors because everyone owns cattle and has a collective interest in the managed pools and the availability of water throughout the year.

It strikes me, then, that this is a principle of the human-altered world—at its foundation, an ethical matter of mindfulness about how intimately we are all connected.

I've come to see the Anthropocene, therefore, not as debate about a new geological era but rather as a way of thinking—a way of thinking about our identity, and what it will mean to be human in the future. And so, our "thought experiment" here focuses less on specific problems, less on piecemeal solutions to the harm people may induce, and more on the principles that may guide meaningful pathways as we continue to alter the world and ourselves.

There is much to appreciate in a single origin that nurtures human identity as a species. The effect of a planetary, one-humanity narrative is to foster a sense of collective identity, the value of collective well-being, and a sense of shared responsibility for that well-being.

The positive paths we create in this Age of Humans will not be reached by a total consensus (that would not be "human" of us at all!). Yet in seeking meaningful futures, people must feel included in the community, national and global conversations. Inclusion can enable people to listen, reflect and act coherently even if actions are an expression of our inherent diversity.



Homo heidelbergensis lived in a social group that controlled fire, built shelters, hunted animals and shared food with each other. (Bronze artist: John Gurche)

We need to get over our mourning for an ancient concept of nature as pristine, eternal if only people would leave it alone. This idea defines nature as something that exists beyond where people live, and thus is now largely invisible, inaccessible, and irrelevant to most people. Such an unchanging, original natural world is a misreading of nature. And it draws from the mistaken assumption that humans are separate from it by special dominion and mastery of the environment. This old myth offers none of the insights and understandings on which human lives depend as part of physical and biotic systems across the planet.

One of the most important principles to consider is resilience, or adaptability—a dynamic process. It means the capacity to adjust by processes of change and evolution. It is critical, however, to distinguish resilience from sustainability, another Anthropocene concept. In defining what we want the world to be, I think we all seek to sustain "the world that's familiar to us"—the world as we see it. Yet an intended future will need to be defined in far more dynamic, ever-changing terms. Each decade will comprise a newly altered world. Understandings and hopes will need to be framed in ways we cannot begin to see. Every new generation will live in a new Anthropocene.

Much as we who 'came of age' and grew up in the 1960s and 70s built our lives around new and perhaps radical assumptions regarding personal liberties and equalities that proved troubling to many in previous generations, so should we be mindful to find inspiration and celebration, rather than threat, as future generations experiment and define new expectations built on the principle of resilience rather than on our desire to preserve the powerful pull of the world as we see it and demand its preservation.

Certain definitions of sustainability are simply too static, seeking to stabilize what already exists, and to preserve the status quo, although it is unclear whose status quo should be preserved. World climate and other nonhuman systems are unpredictable enough; human activities will continue to add new unpredictable effects. The combination will challenge our adaptability. This is one of the deep-time principles of human origins, and it is likely to continue as a principle of human origins. It seems wise not to anticipate a future that's any different at least in this regard.

We can certainly agree every person has some stake in the health, abundance and transformations of the world around us. Planning for purposeful, beneficial outcomes will need to be in touch with the realities of human alteration of the environment, mismanagement, species loss and the miseries inherent in the range of human impulses and conflicts. And so, when it comes to building principles for living in the Anthropocene, there is certainly a need for people to become morally aroused and activated, with a deep sense of personal responsibility that will stretch us beyond self-interest.



Like a modern human child, the curious two-year old *Homo neanderthalensis* learns as he watches. (Bronze artist: John Gurche)

In this light, I can suggest certain qualities that will contribute to a moral stance in the Anthropocene: Universality, inclusion, empathy, reciprocity, humility, connection to something larger than ourselves, our embeddedness in nature, a union of anthropocentric and biocentric thinking, which combines reasoning beneficial to both the human and nonhuman realms. There are more qualities that could be noted, and a conversation about each would require a lot of discussion. Yet I am convinced such qualities must become part of the shared social project of our new era. Inclusion is the right of people to participate in a decision, which is linked to justice. Humility is the opposite of a certain sense of the word "dominion." Embeddedness in nature is seeing ourselves as evolved as part of the natural world rather than separated from it. Empathy and reciprocity arise from taking the perspective of others.

Critical to raising resilience, and thus to life in the Anthropocene, is what I call the Moral Responsibility Dilemma. This dilemma arises from the fact that humans are a global phenomenon, and we are packed together in closer proximity than ever before. It goes something like this: In a situation where people perceive that self-restraint is important in using a resource, or in solving a particular environmental problem, yet at the same time, it's also perceived that others (others in the community, other nations, and so on) do not share a similar belief or commitment, what then develops is a sense of an unequal moral investment. When that occurs, personal responsibility goes out the window, and no restraint or solution is acted on.

Solving this dilemma and its impact on the psychology of human action will be a major project of the Anthropocene. It will be impossible to make progress on this dilemma without a planetary and one-species narrative reminding us we're all in it together in terms of solving the continuing challenges of resilience and responsibility.

About Rick Potts



Paleoanthropologist Dr. Rick Potts heads the Human Origins Program at the Smithsonian’s National Museum of Natural History. In partnership with the Kenya Museums, Potts leads ongoing excavations in southern and western Kenya. His research focuses on how environmental instability has affected human evolution and our evolutionary adaptations. He is the curator of the Smithsonian’s "Hall of Human Origins" and the traveling exhibit “Exploring Human Origins.” He also authored the exhibit companion book *What Does It Mean To Be Human?*

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